

EndoPrior-GS: Dynamic Endoscopic Reconstruction with a Joint Texture Prior

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Abstract. Dynamic endoscopic reconstruction is fundamental to robotic surgery and computer-assisted interventions. While 3D Gaussian Splatting (3DGS) realises real-time rendering, its application to deformable intraoperative environments remains constrained by phantom geometry and varying illuminations. To address these limitations, we introduce EndoPrior-GS, a novel pipeline that explicitly couples frame-extracted vision heuristics and estimated depth maps. EndoPrior-GS derives a joint texture prior from a tool-filtered valid tissue mask, a non-specular photometric filter, and anatomical structural salience, yielding a probability map that guides primitive initialisation and subsequent density control. The prior is further extended to the temporal domain through a texture-aware term that dynamically weighs pairwise primitive contributions during training. We conduct extensive experiments on benchmark datasets EndoNeRF and SCARED, and the obtained results show that our method EndoPrior-GS reduces Flow Error by 27.7% and 25.8% over the representative approaches while preserving competitive rendering quality and real-time rendering speed.

Keywords: Dynamic endoscopic reconstruction · 3D Gaussian Splatting · Prior-guided optimisation · Surgical scene reconstruction

1 Introduction

In recent years, the development of computer-assisted interventions, such as diagnostic endoscopy, has shifted towards generating real-time rendering and high-fidelity 3D reconstruction of the tissue field to enhance intraoperative navigation and visual interpretation of complex, non-rigid anatomical structures [3, 12, 17, 19]. While continuously implicit representations such as Neural Radiance Fields (NeRF) have shown promise for this task, the considerable computational overhead has hindered true real-time rendering [15, 23].

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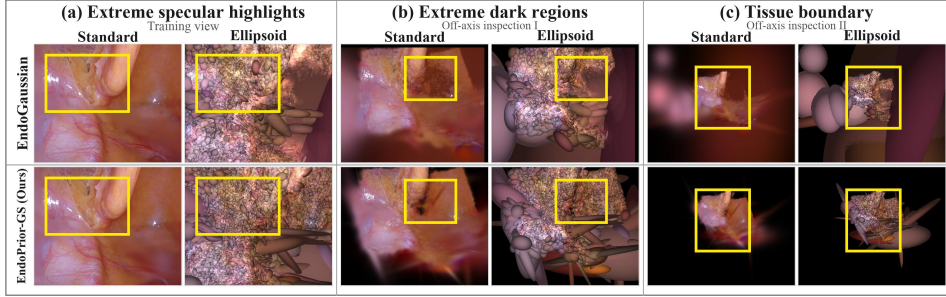


Fig. 1: Comparison of rendering results in challenging endoscopic regions. (a) Extreme specular highlights, (b) extreme dark regions, and (c) tissue boundaries show the three cases targeted by EndoPrior-GS (Ours).

The emergence of 3D Gaussian Splatting (3DGS) has fundamentally transformed novel view synthesis, enabling high-speed synthesis via either explicit point-based representations [11, 29, 30] or pixel-wise re-projection [4, 5, 21, 26], coupled with tile-based rasterisation. However, adapting vanilla 3DGS to dynamic endoscopic sequences is challenging. Standard 3DGS relies heavily on Structure-from-Motion (SfM) frameworks like COLMAP [18] for initial sparse point cloud generation, which usually fails in endoscopy because of the uniform, featureless textures of internal organs and the severe spatial constraints of monocular camera trajectories [7, 13].

The depth-reprojection pipelines [9, 13, 25], on the other hand, leverage monocular depth estimators to guide the initialisation of Gaussian primitives. Whilst computationally expedient, these approaches treat every pixel uniformly. Specifically, depth estimation cannot precisely capture geometry on highly reflective instruments, preserve sharp anatomical boundaries, and distinguish ambiguous regions under varying illumination as illustrated in Fig. 1. Furthermore, when dynamic instruments pass across the tissue bed, both point-based and depth reprojection-based approaches struggle to maintain long-range inter-frame consistency, as the temporal correspondence of underlying tissue becomes non-rigid and deformable. The Flow Error as evidence for different benchmarked approaches is shown in Fig. 2.

To overcome these limitations, we present a novel pipeline, EndoPrior-GS, that explicitly couples the depth maps with frame-wise vision heuristics to guide splatting with structural texture information. Firstly, we construct a probability map by taking the product of a binary tool-filtered valid tissue mask, a non-specular photometric filter, and anatomical structural salience. This probability map then serves as a filter and priority guide that can be formulated and accumu-

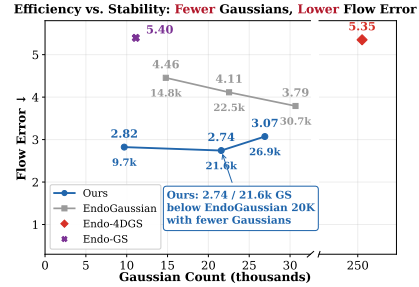


Fig. 2: Efficiency and Stability. EndoPrior-GS (Ours) achieves lower Flow Error with fewer Gaussians.

lated into a persistent texture-prior score to dynamically regulate the entire splatting process. We further introduce a texture-aware temporal regularisation term that can be imposed on the original 3DGS optimisation pipeline to dynamically assign different strengths to local neighbourhood primitive relations to enforce inter-frame consistency.

The principal contributions are summarised as follows:

- We propose a training-free **Joint Texture Prior**, derived from frame-wise visual heuristics, to provide an explicit prior-based mechanism for guiding Gaussian primitive evolution.
- We develop a **Prior-Guided Initialisation and Density Control** strategy that regulates Gaussian initialisation and density updates, reducing redundant growth in geometrically distorted regions.
- We introduce a **Texture-Aware Temporal Consistency** term that dynamically assigns different strengths to local neighbourhood primitive relations to enforce inter-frame consistency.

We conducted experiments to show that EndoPrior-GS reduces Flow Error by 27.7% and 25.8% on EndoNeRF and SCARED, respectively, while improving temporal consistency and suppressing redundant and unstable Gaussian growth.

2 Related Work

2.1 Endoscopic Neural Reconstruction

Dynamic endoscopic reconstruction is hindered by non-rigid tissue deformation, restricted camera trajectories, tool occlusion, specular reflections, motion blur, and ambiguous or weak tissue texture [6, 14, 23, 30]. Neural rendering methods have adapted radiance-field and neural-surface representations to surgical scenes. EndoNeRF [23] reconstructs deformable tissues with neural rendering, mask-guided ray sampling, and stereo depth priors, while EndoSurf [27] combines deformation, signed-distance, and radiance fields for dynamic surface reconstruction. These implicit formulations encode scene support as continuous fields queried along rays, requiring costly ray sampling and leaving no persistent local elements for direct refinement or explicit temporal constraints.

2.2 Dynamic Endoscopic Reconstruction with 3DGS

Explicit 3D Gaussian Splatting offers a more controllable alternative for efficient dynamic endoscopic reconstruction [11]. Existing methods mainly incorporate auxiliary cues at specific stages.

Several methods address the initial Gaussian construction by re-projection from depth maps. EndoGaussian [13] proposes Holistic Gaussian Initialisation (HGI), where pixels from input frames are re-projected using estimated depth maps. SurgicalGaussian [25] follows the same general direction by using depth priors. Endo-4DGS [9] uses pseudo-depth maps for monocular Gaussian initialisation and applies confidence-guided depth learning during optimisation. These

methods mainly apply depth and mask cues to construct or supervise the initial Gaussian set; subsequent primitive growth and pruning remain governed by standard 3DGS optimisation criteria.

Other works improve endoscopic Gaussian deformation, surface geometry, or appearance modelling. Endo-GS [30] utilises depth-guided supervision and surface-aligned regularisation to improve deformable tissue reconstruction under tool occlusion. SurgicalGaussian [25] imposes consistency losses on Gaussian position and covariance to regularise local motion. Surgical Gaussian Surfels [20] models tissue geometry with Gaussian surfels to avoid unrestricted volumetric scaling of standard 3DGS. Endo-4DGX [8] incorporates illumination embeddings for illumination-adaptive colour prediction. Although these methods are effective, they mostly address specific failure modes through targeted supervision, regularisation, geometric modelling, or appearance conditioning.

Rather than treating endoscopy-specific challenges merely as isolated terms in the training objective or as stand-alone correction modules, we re-examine the optimisation flow of 3DGS. The proposed EndoPrior-GS explicitly couples frame-extracted vision heuristics into the Gaussian splatting optimisation pipeline throughout training.

3 Method

3.1 Overview

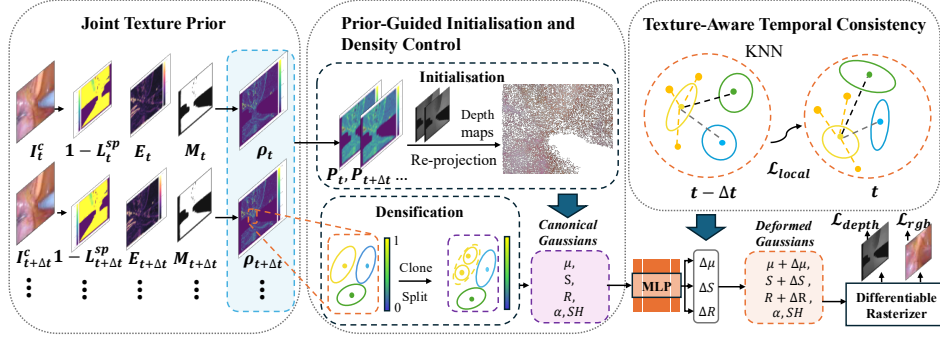


Fig. 3: The proposed EndoPrior-GS framework.

The core pipeline of the proposed EndoPrior-GS is illustrated in Fig. 3. Specifically, we construct a unified texture prior that operates as a training-free heuristic mechanism. Computed directly from low-level image cues, this prior circumvents the need for heavy feature-extraction networks commonly found in standard approaches. The joint texture prior, alongside the raw input frame, is first normalised into an initialisation sampling distribution that adaptively determines whether a 3D Gaussian primitive should be initialised at a given pixel location. Furthermore, we formulate a texture-prior scoring function to

dynamically govern the lifecycle of the Gaussians during the subsequent densification and pruning stages. Lastly, a texture-aware temporal regularisation term is integrated into the optimisation flow to enforce spatio-temporal consistency, ensuring stable and high-fidelity reconstruction of highly deformable dynamic endoscopic scenes.

3.2 Joint Texture Prior

We construct a joint texture prior to address the scenarios where standard 3DGS struggles when applied to dynamic endoscopic scenes, which are pervasively corrupted by instrument occlusions or background noise, specular reflections and lack of distinctive textures. These adverse factors often trigger floating artefacts, as illustrated in Fig. 1.

To mitigate this, we propose a joint texture prior ρ_t as a pixel-wise spatial probability map over the image domain Ω_t , which denotes the discrete 2D pixel-coordinate space of the image frame t . For any pixel location $\mathbf{u} = (u, v) \in \Omega_t$, $\rho_t(\mathbf{u})$ is computed directly from low-level image cues:

$$\rho_t(\mathbf{u}) = M_t(\mathbf{u})(1 - L_t^{\text{sp}}(\mathbf{u}))E_t(\mathbf{u}), \quad (1)$$

where $M_t(\mathbf{u}) \in \{0, 1\}$ is the value of the binary tool-filtered valid tissue mask at pixel $\mathbf{u}$, $L_t^{\text{sp}}(\mathbf{u}) \in [0, 1]$ is the specular likelihood at $\mathbf{u}$, and $E_t(\mathbf{u}) \in [0, 1]$ is the anatomical structural salience at $\mathbf{u}$.

Specifically, the mask M_t is extracted directly from the dataset, when $M_t(\mathbf{u}) = 1$ indicates usable tissue and $M_t(\mathbf{u}) = 0$ indicates tool or invalid pixels. The reflection intensity map L_t^{sp} is estimated from RGB intensity and saturation cues [1, 16]. The specular likelihood $L_t^{\text{sp}}(\mathbf{u}) \in [0, 1]$ at pixel $\mathbf{u}$ is computed as:

$$L_t^{\text{sp}}(\mathbf{u}) = \max \left\{ \sigma \left(\frac{V_t(\mathbf{u}) - \tau_v}{\gamma_v} \right) \sigma \left(\frac{\tau_s - S_t(\mathbf{u})}{\gamma_s} \right), C_t(\mathbf{u}) \right\}, \quad (2)$$

where $V_t(\mathbf{u}) = \max_c I_t^c(\mathbf{u})$ is the maximum RGB intensity, $I_t^c(\mathbf{u})$ is the intensity of RGB colour channel $c \in \{R, G, B\}$ at pixel $\mathbf{u}$ in frame t , and $S_t(\mathbf{u}) = (V_t(\mathbf{u}) - \min_c I_t^c(\mathbf{u})) / (V_t(\mathbf{u}) + \epsilon)$ is the saturation computed from the RGB channels. C_t is a binary saturated-channel indicator map, and $C_t(\mathbf{u}) \in \{0, 1\}$ is its value at pixel $\mathbf{u}$. Specifically, $C_t(\mathbf{u}) = 1$ when at least two RGB channels at $\mathbf{u}$ exceed τ_o , and $C_t(\mathbf{u}) = 0$ otherwise. $\sigma(\cdot)$ denotes the standard sigmoid activation function, which converts the intensity and saturation response into the continuous interval $[0, 1]$; τ_v and τ_s are the brightness and saturation thresholds, γ_v and γ_s are the temperature parameters controlling the sigmoid transition around τ_v and τ_s .

Concurrently, the structural salience value $E_t(\mathbf{u}) \in [0, 1]$ measures the strength of anatomical edges and texture at pixel $\mathbf{u}$, which provides useful landmarks for tracking non-rigid tissue deformation. To compute this, the RGB frame is converted to a grayscale intensity map I_t . The horizontal and vertical gradients

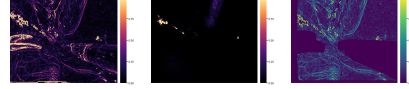


Fig. 4: Joint Texture Prior. E_t , L_t^{sp} , and $\rho_t = M_t \odot (1 - L_t^{\text{sp}}) \odot E_t$.

$G_x(\mathbf{u})$ and $G_y(\mathbf{u})$ at pixel $\mathbf{u}$ are calculated by convolving the image with directional 3×3 Sobel kernels, K_x and K_y :

$$G_x(\mathbf{u}) = (K_x * I_t)(\mathbf{u}), \quad G_y(\mathbf{u}) = (K_y * I_t)(\mathbf{u}), \quad (3)$$

where the kernels are defined as:

$$K_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad K_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}. \quad (4)$$

The unnormalised edge magnitude is obtained via the Euclidean norm $\hat{E}_t(\mathbf{u}) = \sqrt{G_x(\mathbf{u})^2 + G_y(\mathbf{u})^2}$. To bound this structural prior to the continuous range $[0, 1]$, we normalise the edge magnitude using a robust percentile computed over valid-tissue pixels within the image domain Ω_t :

$$E_t(\mathbf{u}) = \min \left\{ \frac{\hat{E}_t(\mathbf{u})}{Q_p(\{\hat{E}_t(\mathbf{x}) \mid \mathbf{x} \in \Omega_t, M_t(\mathbf{x}) = 1\}) + \epsilon}, 1 \right\}, \quad (5)$$

where $Q_p(\cdot)$ is the p -th percentile, and ϵ is a small numerical stabiliser. The cue maps and the resulting joint texture prior are illustrated in Fig. 4.

3.3 Prior-Guided Initialisation and Density Control

The joint texture prior is explicitly leveraged to guide the initialisation, densification, and pruning of Gaussian primitives. This approach fundamentally departs from the standard adaptive density-control operations of 3DGS [11], ensuring that the Gaussian allocation evolves selectively and aligns with the extracted structurally valid tissue regions.

Initialisation: Standard 3DGS initialises primitives using sparse Structure-from-Motion (SfM) [18] point clouds. In endoscopic scenes, this triggers catastrophic failures as points erroneously anchor to transient surgical-tool regions or shifting specular highlights. To overcome this, we leverage the joint texture prior ρ_t to construct an initialisation sampling distribution P_t over the image domain Ω_t at frame t . For each pixel $\mathbf{u} \in \Omega_t$, its sampling probability is defined as:

$$P_t(\mathbf{u}) = \eta_{\text{app}} \frac{\widetilde{M}_t(\mathbf{u})}{\sum_{\mathbf{x} \in \Omega_t} \widetilde{M}_t(\mathbf{x}) + \epsilon} + (1 - \eta_{\text{app}}) \frac{\rho_t(\mathbf{u})}{\sum_{\mathbf{x} \in \Omega_t} \rho_t(\mathbf{x}) + \epsilon}, \quad (6)$$

where ϵ is a small numerical stabiliser, and $\eta_{\text{app}} \in [0, 1]$ is a balancing coefficient. The first term enforces uniform coverage over the valid tissue, whilst the second term prioritises non-specular, structurally rich regions. To prevent initialisation on unstable tool boundaries, the raw mask M_t is conservatively refined via morphological erosion: $\widetilde{M}_t = M_t \ominus B_k$, where $\widetilde{M}_t(\mathbf{u}) = \min_{\mathbf{v} \in B_k} M_t(\mathbf{u} + \mathbf{v})$, with $\ominus$ denoting the erosion operator and B_k representing an elliptical structuring element of size $k \times k$ ($k = 9$).

After sampling a set of initialisation pixels from P_t , we back-project them with the estimated depth maps to form the initial Gaussian point cloud.

Densification: The complex and dynamic nature of endoscopic imaging, including specular highlights, tool boundaries, or motion blur, poses significant challenges for primitive densification, often resulting in severe geometric distortions. We address these vulnerabilities by introducing a persistent texture-prior score $\bar{s}_i$ for each Gaussian that samples ρ_t at its projected image coordinates across frames, which attenuates densification for primitives repeatedly projected onto low-prior regions.

To obtain the persistent texture-prior score $\bar{s}_i$, we first project each deformed Gaussian centre denoted by $\boldsymbol{\mu}_{i,t} \in \mathbb{R}^3$ at frame t onto the image plane as $\mathbf{c}_{i,t} = \pi(\boldsymbol{\mu}_{i,t}; \Pi_t)$, where $\pi(\cdot; \Pi_t)$ is the camera projection function with camera parameters Π_t , and $\mathbf{c}_{i,t} \in \mathbb{R}^2$ is the projected image coordinate.

The texture-prior value of primitive i at frame t is computed as $s_{i,t} = m_{i,t} \mathcal{B}(\rho_t, \mathbf{c}_{i,t})$, where $\mathcal{B}$ denotes bilinear interpolation of ρ_t at $\mathbf{c}_{i,t}$, and $m_{i,t} \in \{0, 1\}$ is the projection-validity indicator that is set to $m_{i,t} = 1$ when $\mathbf{c}_{i,t}$ lies within the image bounds and $M_t(\mathbf{c}_{i,t}) = 1$, and to $m_{i,t} = 0$ otherwise. The texture-prior score $\bar{s}_i$ is then obtained by averaging valid samples:

$$\bar{s}_i = \frac{\sum_t s_{i,t}}{\sum_t m_{i,t} + \epsilon}. \quad (7)$$

The densification, therefore, can be modulated by incorporating the texture-prior score $\bar{s}_i$ derived from Eq. (7), and denoted as:

$$\tilde{g}_i = [\lambda_{\min}^{\text{den}} + (1 - \lambda_{\min}^{\text{den}}) \bar{s}_i] g_i, \quad (8)$$

where g_i is the view-space position-gradient magnitude, and $\lambda_{\min}^{\text{den}} \in [0, 1]$ is the minimum retained densification weight. This modulation attenuates densification triggers from artefact-induced gradients, thereby reducing redundant Gaussian growth around geometrically distorted regions. This modulated densification mechanism is illustrated in Fig. 5, and more visuals can be found in Fig. 9.

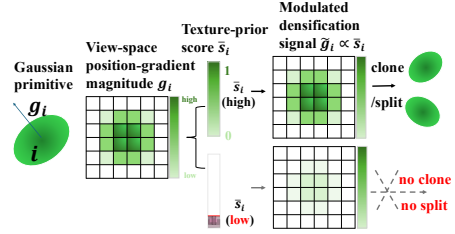


Fig. 5: Prior-Guided Densification. The densification trigger is modulated by the texture-prior score $\bar{s}_i$.

Pruning: Our pruning strategy explicitly integrates the obtained texture-prior score $\bar{s}_i$ alongside standard density-control criteria. Concretely, a primitive needs to satisfy the following criteria: $n_i \geq n_{\min}$, $\bar{s}_i < \tau_{\text{prior}}$, $\bar{\alpha}_i < \tau_{\alpha}$, where $n_i = \sum_t m_{i,t}$ denotes the valid-projection count of Gaussian primitive i , $\bar{\alpha}_i$ is the running mean opacity during training, τ_{prior} and τ_{α} are the primitive texture-prior threshold and the opacity threshold. This criterion targets semi-transparent primitives that are repeatedly projected onto low-prior regions, while avoiding aggressive pruning of newly created or insufficiently sampled Gaussians.

3.4 Texture-Aware Temporal Consistency

While prior-guided initialisation and density control optimise the spatial distribution of Gaussian primitives, spatially heterogeneous anatomical motion may drive the deformation field to produce incoherent local offsets. To address this limitation, we incorporate a texture-aware temporal regularisation term that dynamically adjusts each primitive’s contribution over time and encourages neighbouring Gaussian primitives to move coherently.

Since all frames share a canonical Gaussian set, each primitive has a consistent identity. At frame t , we first construct an eligible primitive set $\mathcal{S}_t$ containing primitives that have valid projections at both $t - \Delta t$ and t , and meet the criteria of having a minimum number of valid projected samples, such that $n_i \geq n_{\min}$. Here, Δt represents the normalised interval between adjacent training frames, and n_i denotes the valid projection count of primitive i . Then, for each eligible primitive $i \in \mathcal{S}_t$, we assign a primitive-specific motion weight w_i^{mot} derived from its texture-prior score $\bar{s}_i$, which is formulated as:

$$w_i^{\text{mot}} = \sigma \left(\frac{\bar{s}_i - \tau_m}{\gamma_m} \right), \quad (9)$$

where $\sigma(\cdot)$ denotes the standard sigmoid activation function that maps the resulting scalar to the continuous interval $[0, 1]$, τ_m is a texture-prior threshold for motion weight, and γ_m represents a temperature scaling parameter.

We further extend this to pairwise primitive relations to ensure local deformation constraints are confined to pairs where both primitives are mutually trustworthy. Specifically, we compute a pairwise spatio-temporal affinity weight $w_{ij,t}$ between primitive i and its neighbour j at frame t , which is defined as:

$$w_{ij,t} = \exp \left(- \frac{\|\boldsymbol{\mu}_{i,t-\Delta t} - \boldsymbol{\mu}_{j,t-\Delta t}\|_2^2}{2\sigma_{\text{loc}}^2} \right) w_i^{\text{mot}} w_j^{\text{mot}}, \quad (10)$$

where σ_{loc} controls the spatial range of the local neighbourhood, and $\boldsymbol{\mu}_{i,t-\Delta t}$ and $\boldsymbol{\mu}_{j,t-\Delta t}$ denote the 3D spatial centres of primitives i and j at the preceding reference frame $t - \Delta t$, respectively.

The proposed texture-aware temporal consistency loss that leverages the above prior-driven weights to ensure a smooth deformation and maintain structural coherence across neighbouring frames is specifically defined as:

$$\begin{aligned} \mathcal{L}_{\text{local}} = & \lambda_{\text{disp}} \frac{\sum_{i \in \mathcal{S}_t} \sum_{j \in \mathcal{N}_K^{(t)}(i)} w_{ij,t} \|\mathbf{d}_{i,t} - \mathbf{d}_{j,t}\|_2}{\sum_{i \in \mathcal{S}_t} \sum_{j \in \mathcal{N}_K^{(t)}(i)} w_{ij,t} + \epsilon} \\ & + \lambda_{\text{dist}} \frac{\sum_{i \in \mathcal{S}_t} \sum_{j \in \mathcal{N}_K^{(t)}(i)} w_{ij,t} h_{\delta}(|\ell_{ij,t} - \ell_{ij,t-\Delta t}|)}{\sum_{i \in \mathcal{S}_t} \sum_{j \in \mathcal{N}_K^{(t)}(i)} w_{ij,t} + \epsilon}, \end{aligned} \quad (11)$$

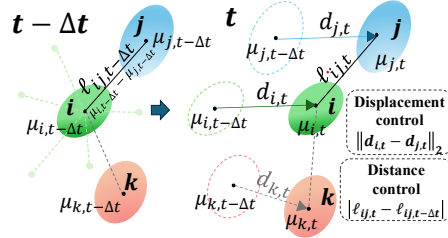


Fig. 6. Temporal Regularisation.

where $\mathcal{N}_K^{(t)}(i)$ is the local neighbourhood of primitive i , containing the K nearest Gaussian primitives in $\mathcal{S}_t \setminus \{i\}$ with distances measured at the preceding reference frame $t - \Delta t$. The adjacent training frame displacements of primitives i and j are $\mathbf{d}_{i,t} = \boldsymbol{\mu}_{i,t} - \boldsymbol{\mu}_{i,t-\Delta t}$ and $\mathbf{d}_{j,t} = \boldsymbol{\mu}_{j,t} - \boldsymbol{\mu}_{j,t-\Delta t}$, respectively. The pairwise Gaussian centre distance of primitive i and j at frame t is defined as $\ell_{ij,t} = \|\boldsymbol{\mu}_{i,t} - \boldsymbol{\mu}_{j,t}\|_2$. The function $h_\delta(\cdot)$ denotes the Huber penalty with transition parameter δ [10], λ_{disp} and λ_{dist} balance the first and second terms, and ϵ is a small constant for numerical stability. The first term reduces isolated primitive motion drift by encouraging neighbouring primitives to have similar adjacent-frame displacements. The second term discourages sudden local stretching or collapse by penalising changes in pairwise Gaussian centre distances. The mechanism is illustrated in Fig. 6.

The training objective of our framework is formulated as a linear combination of baseline objectives and our proposed texture-aware constraint. Formally,

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rgb}} + \lambda_{\text{depth}}\mathcal{L}_{\text{depth}} + \lambda_{\text{tv}}\mathcal{L}_{\text{tv}} + \lambda_{\text{local}}\mathcal{L}_{\text{local}}, \quad (12)$$

where $\mathcal{L}_{\text{rgb}}$, $\mathcal{L}_{\text{depth}}$, and $\mathcal{L}_{\text{tv}}$ represent the masked RGB colour loss, depth loss, and total variation (TV) loss introduced by EndoGaussian [13], respectively; the scaling coefficients λ_{depth} , λ_{tv} , and λ_{local} are hyperparameters that balance the contribution of each respective penalty during the joint optimisation flow.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate our method on two publicly released dynamic endoscopic reconstruction benchmarks: **EndoNeRF** [23] and **SCARED** [2].

The **EndoNeRF** dataset [23] consists of six stereo video sequences captured from DaVinci robotic surgical procedures, totalling 807 frames. Each sequence has a resolution of 512×640 and exhibits significant non-rigid tissue deformation and tool occlusion. Following prior work [9, 13], we evaluate on two public cases (*pulling* and *cutting*) using a 7:1 train-test split.

The **SCARED** dataset [2] consists of stereo RGB-D endoscopic sequences captured from porcine cadaver abdominal anatomies using a DaVinci endoscope. We evaluate on datasets 1, 2, 3, 6, and 7. Following the standard protocol used in previous endoscopic reconstruction works [13, 27], we use a fixed temporal interval split on each sequence into training and testing frames with a 7:1 ratio.

Metrics. We assess rendering quality using standard image metrics: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM) [24], and Learned Perceptual Image Patch Similarity (LPIPS) [28]. Depth accuracy is measured using root mean square error (RMSE) over pixels with valid depth annotations. To evaluate temporal consistency, we report a RAFT-based Flow Error [22] between the rendered and ground-truth video sequences within valid tissue regions. We also report the Gaussian primitive count N_{GS} after training and rendering speed in frames per second (FPS) to measure rendering efficiency.

A rendered-depth temporal instability diagnostic is reported in Sec. 4.4 to quantify adjacent-frame depth variation. The diagnostic is computed for an adjacent frame pair $(t - \Delta t, t)$ as $\mathcal{I}_t = \frac{1}{|\Omega_{t-\Delta t, t}^{\text{val}}|} \sum_{\mathbf{u} \in \Omega_{t-\Delta t, t}^{\text{val}}} |\hat{D}_t(\mathbf{u}) - \hat{D}_{t-\Delta t}(\mathbf{u})|$, where $\hat{D}_t(\mathbf{u})$ denotes the rendered depth in frame t at pixel $\mathbf{u}$. The evaluation domain is $\Omega_{t-\Delta t, t}^{\text{val}} = \left\{ \mathbf{u} \in \Omega_t \mid M_{t-\Delta t}(\mathbf{u}) = M_t(\mathbf{u}) = 1, \hat{D}_{t-\Delta t}(\mathbf{u})\hat{D}_t(\mathbf{u}) > 0 \right\}$, where $M_{t-\Delta t}(\mathbf{u}) = M_t(\mathbf{u}) = 1$ indicates that pixel $\mathbf{u}$ is retained by the binary tool-filtered valid tissue masks in both frames, and $\hat{D}_{t-\Delta t}(\mathbf{u})\hat{D}_t(\mathbf{u}) > 0$ indicates valid rendered depth values in both frames.

Implementation details. All experiments follow the training schedule as EndoGaussian [13], with 1,000 coarse-stage iterations and 3,000 fine-stage iterations, and are conducted on an NVIDIA RTX 6000 Ada Generation GPU.

In Sec. 3.2, the specular likelihood in Eq. 2 uses $\tau_v = 0.85$, $\tau_s = 0.35$, $\gamma_v = 0.04$, $\gamma_s = 0.08$, and saturated-channel threshold $\tau_o = 0.92$. For Eq. 5, we set percentile $p = 99$. In Sec. 3.3, we set the initialisation balancing coefficient in Eq. 6 to $\eta_{\text{app}} = 0.45$. The texture-prior score $\bar{s}_i$ is updated every 100 iterations, following 3DGS [11]. Prior-guided densification applies Eq. 8 with $\lambda_{\text{min}}^{\text{den}} = 0.35$. During pruning, we set $\tau_{\text{prior}} = 0.2$, $n_{\text{min}} = 20$, opacity threshold $\tau_\alpha = 0.0075$. In Sec. 3.4, the weight in Eq. 9 is initialised with $\tau_m = 0.45$ and $\gamma_m = 0.08$. We set $\lambda_{\text{dist}} = 0.2$, $\lambda_{\text{disp}} = 1.0$, $K = 8$ and Huber transition parameter $\delta = 0.01$ in Eq. 11, respectively. The texture-aware temporal regularisation term is weighted by $\lambda_{\text{local}} = 10^{-6}$ in Eq. 12. We use fixed hyperparameters across all sequences.

4.2 Comparison with Representative Methods

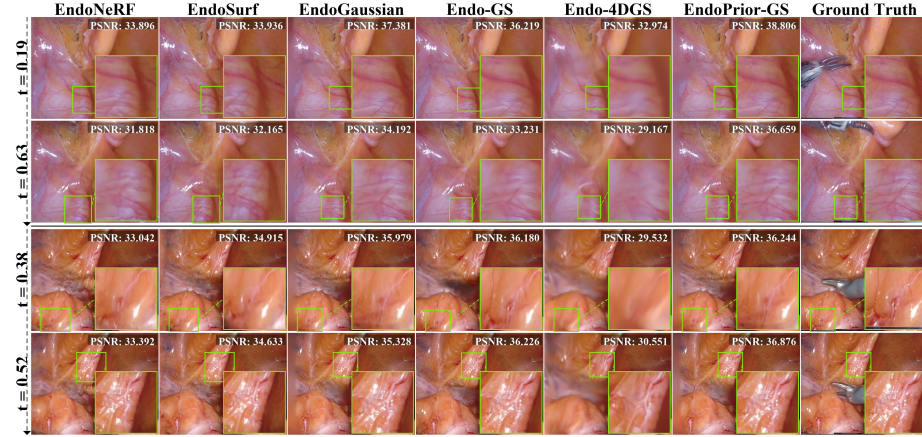


Fig. 7: Qualitative comparison with representative endoscopic reconstruction methods. Results are shown on selected *pulling* and *cutting* sequences. Rows denote timestamps and columns denote methods or ground truth.

Official implementations: ¹ EndoNeRF [GitHub](#); ² EndoSurf [GitHub](#); ³ Endo-GS [GitHub](#); ⁴ Endo-4DGS [GitHub](#); ⁵ EndoGaussian [GitHub](#).

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	RMSE $\downarrow$	Flow Error $\downarrow$	FPS $\uparrow$
EndoNeRF						
EndoNeRF [23] ¹	32.039	0.925	0.081	—	—	—
	27.519	0.933	0.128	2.522	5.421	0.14
EndoSurf [27] ²	34.992	0.954	0.113	1.000	—	—
	35.004	0.845	0.243	2.584	5.284	12.84
Endo-GS [30] ³	37.935	0.966	0.034	—	—	—
	36.710	0.955	0.067	2.477	5.396	76
Endo-4DGS [9] ⁴	37.205	0.957	0.038	—	—	—
	35.353	0.924	0.104	2.686	5.369	15
EndoGaussian [13] ⁵	37.849	0.963	0.054	—	—	—
	37.322	0.958	0.060	2.482	3.791	279
EndoPrior-GS (Ours)	38.071	0.967	0.071	2.260	2.742	305
SCARED						
EndoNeRF [23] ¹	—	—	—	—	—	—
	24.345	0.768	0.313	2.723	5.894	0.06
EndoSurf [27] ²	23.637	0.794	0.351	—	—	—
	25.020	0.802	0.356	2.501	5.401	2
Endo-GS [30] ³	—	—	—	—	—	—
	27.786	0.781	0.245	2.548	4.867	76
Endo-4DGS [9] ⁴	—	—	—	—	—	—
	30.908	0.807	0.167	4.897	4.905	17
EndoGaussian [13] ⁵	27.042	0.827	0.267	—	—	—
	27.070	0.744	0.249	2.675	5.232	269
EndoPrior-GS (Ours)	31.089	0.856	0.196	2.490	3.609	292

Table 1: Quantitative comparison on EndoNeRF [23] and SCARED [2]. Grey cells denote original-paper values, followed by our reproduced results under a unified protocol. Red superscripts indicate official GitHub links. Best and second-best results are highlighted in red and orange.

Table 1 compares EndoPrior-GS (Ours) with representative endoscopic reconstruction methods from two categories: (i) neural reconstruction methods, including **EndoNeRF** [23] and **EndoSurf** [27]; and (ii) Gaussian-based methods, including **EndoGaussian** [13], **Endo-GS** [30], and **Endo-4DGS** [9]. EndoPrior-GS (Ours) consistently achieves more stable reconstruction in the temporal domain, as reflected by lower Flow Error. On EndoNeRF [23]: Compared with EndoGaussian [13], EndoPrior-GS (Ours) reduces Flow Error from 3.791 to 2.742, corresponding to a **27.7%** relative reduction, while maintaining competitive rendering quality and speed. On SCARED [2]: Compared with Endo-GS [30], the strongest baseline in terms of Flow Error, EndoPrior-GS (Ours) reduces Flow Error from 4.867 to 3.609, giving a **25.8%** relative reduction. Fig. 7 further shows that EndoPrior-GS preserves more continuous tissue structures in challenging regions, consistent with the quantitative improvements.

Budget	Method	$N_{\text{GS}}\downarrow$	PSNR $\uparrow$	RMSE $\downarrow$	Flow Error $\downarrow$	FPS $\uparrow$
10K	EndoGaussian [13]	14.8K	36.81	2.97	4.46	307
	EndoPrior-GS	9.7K	36.87	2.47	2.82	310
20K	EndoGaussian [13]	22.5K	37.14	2.80	4.11	295
	EndoPrior-GS	21.6K	38.07	2.26	2.74	305
30K	EndoGaussian [13]	30.7K	37.32	2.48	3.79	280
	EndoPrior-GS	26.9K	38.12	2.27	3.07	300

Table 2: Efficiency and stability comparison under matched initialisation budgets. The budget denotes the nominal initialisation setting, while N_{GS} is the Gaussian count after training.

4.3 Efficiency and Stability under Matched Initialisation Budgets

Table 2 evaluates EndoPrior-GS (Ours) and EndoGaussian [13] under matched initialisation budgets. This matched-budget analysis decouples the observed temporal stability gain from Gaussian count. Across the 10K, 20K, and 30K initialisation budgets, EndoPrior-GS achieves lower Flow Error with fewer final Gaussians, indicating that the improvement arises independently of increased primitive capacity.

4.4 Analysis of Temporal Consistency

Beyond the RAFT-based Flow Error, we report the average rendered-depth temporal instability as a complementary diagnostic of temporal stability. We emphasise that these metrics evaluate the consistency of rendered sequences within valid tissue regions. Table 3 reports diagnostic values averaged over all adjacent frame pairs. EndoPrior-GS (Ours) reduces the diagnostic from 0.4459 to 0.3786 on *pulling* sequence and from 0.1166 to 0.1071 on *cutting* sequence, corresponding to 15.1% and 8.1% reductions, respectively. Fig. 8 qualitatively confirms lower local adjacent-frame depth variation in the highlighted tissue regions. The reduction map is EndoGaussian [13] minus EndoPrior-GS (Ours), where positive values indicate lower variation for EndoPrior-GS (Ours). Reported means in Fig. 8 are computed only within the zoomed regions.

Seq.	EndoG. [13]	EndoPrior-GS	Reduction
Pulling	0.4459	0.3786	15.1%
Cutting	0.1166	0.1071	8.1%

Table 3: Average rendered-depth temporal instability diagnostic.

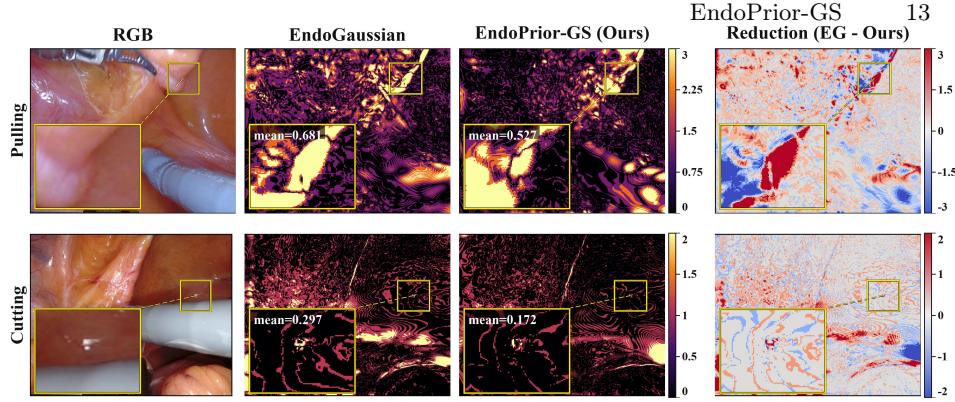


Fig. 8: Comparison of rendered-depth temporal instability diagnostic maps. Heat-maps visualise representative pixel-wise adjacent-frame depth variation.

4.5 Analysis of Primitive Distribution

Variant	$N_{GS} \downarrow$	$N_{low-prior}$	$N_{low-\alpha} \downarrow$	$N_{large-scale} \downarrow$	Flow Error $\downarrow$
Baseline	22.5k	19.9k	262.0	5.0	4.112
w/o Texture-aware temp. reg.	21.5k	19.9k	110.3	0.7	3.005
EndoPrior-GS (Ours)	21.6k	19.9k	105.0	0.0	2.742

Table 4: Quantitative primitive count diagnostic analysis. Counts use fixed post-hoc thresholds after training. The baseline denotes EndoGaussian [13] under 20K initialisation budgets.

The proposed EndoPrior-GS framework suppresses artefact-driven redundant Gaussian growth. Table 4 shows that all variants have comparable final Gaussian counts N_{GS} (21.5K–22.5K) and identical low-prior primitive counts $N_{low-prior}$ (19.9K), measured with $\bar{s}_i < 0.2$ after at least 20 valid observations. Under these comparable counts, EndoPrior-GS (Ours) reduces low-opacity primitives $N_{low-\alpha}$ with opacity < 0.01 from 262.0 to 105.0, and large-scale primitives with normalised-coordinate scale > 40 from 5.0 to 0.0. These diagnostics indicate that the proposed prior-guided strategy does not simply remove low-prior primitives. Instead, it suppresses opacity-ineffective and over-scaled primitive growth while maintaining comparable projected primitive coverage over valid tissue regions. Fig. 9 explains this behaviour qualitatively. Compared with EndoGaussian [13], EndoPrior-GS reduces redundant and unstable growth around surgical-tool and tissue boundary areas.

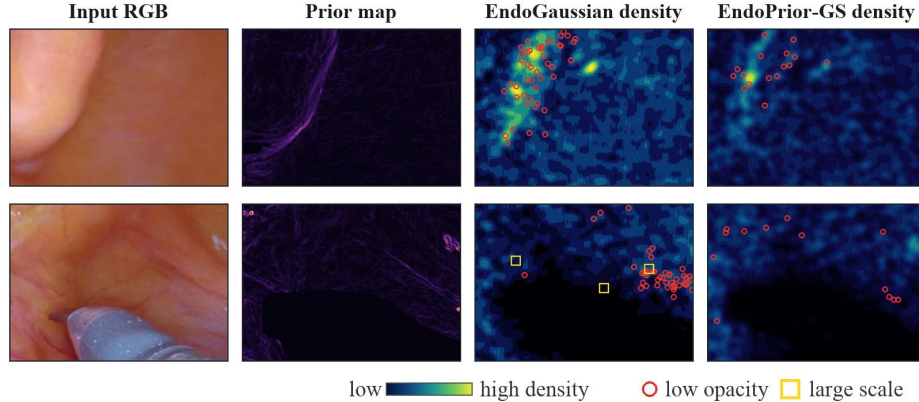


Fig. 9: Qualitative primitive distribution analysis. We show input RGB, joint texture prior ρ_t , and projected Gaussian density, with diagnostic markers computed using the thresholds in Table 4.

4.6 Ablation Study

Texture-aware temporal regularisation ablation. Table 5 isolates the texture-aware temporal regularisation term in Sec. 3.4. The baseline gives the highest Flow Error of 4.112, whereas unweighted temporal regularisation reduces it to 3.035, confirming the benefit of temporal constraints. Introducing pairwise weighting further lowers Flow Error to 2.944, even without pairwise Gaussian centre distance control, indicating that the texture-aware mechanism strengthens adjacent-frame primitive displacement control. The complete formulation achieves the best PSNR, RMSE, and Flow Error, showing that the terms jointly improve temporal consistency.

Variant	Pair. w.	Disp. c.	Dist. c.	PSNR $\uparrow$	RMSE $\downarrow$	Flow Error $\downarrow$
Baseline	$\times$	$\times$	$\times$	37.147	2.804	4.112
w/o Pair. w.	$\times$	$\checkmark$	$\checkmark$	37.057	2.785	3.035
w/o Dist. c.	$\checkmark$	$\checkmark$	$\times$	37.184	2.797	2.944
w/ Texture-aware temp. reg.	$\checkmark$	$\checkmark$	$\checkmark$	37.250	2.575	2.820

Table 5: Ablation study of texture-aware temporal regularisation. “Disp c.” denotes adjacent training frame displacements of primitives control, and “Dist c.” denotes pairwise Gaussian centre distance control over time. “Pair. w.” indicates pairwise spatio-temporal affinity weight.

Components ablation. Table 6 evaluates each component under the matched 20K initialisation budget. Prior-guided initialisation, density control, and texture-aware temporal regularisation each reduce Flow Error relative to the baseline. The full model achieves the best Flow Error, PSNR, and RMSE, indicating that the gains come from propagating the joint texture prior through multiple stages of primitive evolution, rather than from a single post-hoc regularisation term.

5 Conclusion

This paper presents EndoPrior-GS, a novel dynamic endoscopic Gaussian reconstruction framework that uses a heuristic and training-free joint texture prior

Variant	Components				Metrics			
	Pri. init.	Pri. den.	Pri. prune	Temp. reg.	PSNR $\uparrow$	RMSE $\downarrow$	Flow Er. $\downarrow$	$N_{GS}\downarrow$
Baseline	$\times$	$\times$	$\times$	$\times$	37.147	2.804	4.112	22.5k
+ Pri. init.	$\checkmark$	$\times$	$\times$	$\times$	37.206	2.727	3.098	24.0k
+ Pri. den. con.	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	37.088	2.479	3.005	21.5k
+ Tex. temp. reg.	$\times$	$\times$	$\times$	$\checkmark$	37.250	2.575	2.820	21.1k
Full model	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	38.071	2.260	2.742	21.6k

Table 6: Ablation study of the components. All variants are evaluated under the matched 20K budget.

to guide Gaussian primitive evolution. We further extend the texture prior to the temporal domain through a texture-aware term that dynamically weighs pairwise primitive contribution during training. Extensive experiments show that EndoPrior-GS reduces Flow Error by 27.7% and 25.8% on EndoNeRF and SCARED, respectively. For temporal evaluation, Flow Error is computed from RAFT-estimated image-plane optical flow, as dense 3D tissue-motion ground truth is unavailable in current benchmarks; we therefore use it together with an adjacent-frame rendered-depth diagnostic as supporting evidence for rendered temporal stability and consistency. The current prior is derived from low-level image cues and assumes available tool-filtered valid tissue support. It may become less discriminative under pronounced blur, over-saturation, dense smoke occlusion, highly ambiguous tissue texture, or noisy predicted masks.

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